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Descriptor-Injected Cross-Modal Learning: A Systematic Exploration of Audio–MIDI Alignment via Spectral and Melodic Features

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Abstract

Cross-modal retrieval between audio recordings and symbolic music representations (MIDI) remains challenging due to the fundamental modality gap between continuous waveforms and discrete event sequences. We present a systematic exploration of *descriptor injection*—the practice of augmenting modality-specific encoders with hand-crafted domain features—as a mechanism for bridging this gap. Through a three-phase experimental campaign spanning 13 descriptor×mechanism combinations, 6 architectural families, and 3 training schedules, we identify a configuration (D4A4) that reaches a mean S=84.0% ± 2.7 pp across five independent training seeds, a +8.8 percentage point improvement over the descriptor-free baseline. Causal ablation reveals that the audio-side descriptor (A4, octave-band energy dynamics) is entirely responsible for this improvement in the top dual models (D4A4, D4-A4R): zeroing A4 collapses performance by 75–78 pp, while ablating the MIDI-side descriptor (D4, local pitch intervals) changes S by only −0.4 to +0.6 pp despite improving training dynamics. We propose *reverse cross-attention*, a mechanism where descriptor tokens serve as queries attending to encoder features, achieving 163× fewer attention operations than the standard formulation while remaining competitive. Centered Kernel Alignment (CKA) analysis reveals that descriptors double the representational alignment between audio and MIDI transformer layers (0.435 → 0.794, +82%), suggesting that the mechanism of improvement is cross-encoder representational convergence rather than simple feature injection. Perturbation analysis shows that high-frequency octave bands (750–12000 Hz) carry the majority of the discriminative signal, and that this information is encoded non-linearly ($|r| < 0.05$ correlations with output). All experiments use the MAESTRO v3.0.0 piano dataset with a structured evaluation protocol controlling for composer and piece similarity.

1 Introduction

Music exists simultaneously in multiple modalities: as audio waveforms captured by microphones, as symbolic sequences encoded in MIDI files, as visual score images, and as cognitive representations in the listener's mind. The ability to align representations across these modalities—particularly between audio and symbolic music—has direct applications in music retrieval, score following, automatic transcription evaluation, and musicological analysis.

The fundamental challenge of audio–MIDI alignment lies in the *modality gap*: audio is a continuous-time signal shaped by instrument timbre, room acoustics, performer expression, and recording conditions, while MIDI is a discrete sequence of pitch, velocity, and duration events that abstracts away all acoustic detail. Traditional approaches bridge this gap through hand-crafted intermediate

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representations—chromagrams [Ellis, 2007, Ewert and Müller, 2012], audio fingerprints [Wang, 2003], or dynamic time warping on spectral features [Hu et al., 2003]—but these require extensive domain engineering and are brittle to variations in performance style and recording quality.

Recent advances in self-supervised audio representation learning [Baevski et al., 2020, Li et al., 2024] and cross-modal contrastive objectives [Bardes et al., 2022, Radford et al., 2021, Wu et al., 2023] offer a principled alternative: learn a shared embedding space where matched audio–MIDI pairs are close and unmatched pairs are distant. However, naïve application of contrastive learning to audio–MIDI pairs faces a bootstrapping problem: the modality gap is so large that early training gradients are noisy and convergence is slow.

Beyond these practical applications, the question of whether audio and symbolic music share discoverable structural invariants is of broader interest. Frequency-based descriptors—capturing spectral dynamics, harmonic interval patterns, or energy distribution across octave bands—offer a tractable way to test whether certain relational features can serve as cross-modal bridges. If such descriptors prove effective, this would suggest that some aspects of musical information remain legible across modalities through frequency-structured relations rather than through learned abstractions alone.

Hypothesis. We hypothesize that *domain-specific descriptors*—hand-crafted features capturing known invariances within each modality—can serve as a bridge during cross-modal learning. Specifically, spectral dynamics descriptors for audio and melodic interval descriptors for MIDI may provide the encoder with structured prior information that accelerates the discovery of cross-modal correspondences.

Approach. We conduct a systematic three-phase exploration:

1. **Screening** (5-epoch short-horizon): 13 descriptor$\times$injection mechanism combinations, plus 11 alternative architectures across 3 families (FiLM, Mixture-of-Experts, Third Tower).
2. **Confirmation** (30–60 epochs): The top candidates are trained to convergence with optimized learning rate schedules.
3. **Scientific validation** (Gate 5B): A battery of 8 tests—causal ablation, transposition invariance, CKA/RSA alignment, perturbation sensitivity, linear probing, and embedding visualization—applied to the 4 canonical models.

Contributions.

1. A comprehensive taxonomy of 13 descriptor$\times$mechanism combinations for cross-modal music learning, with systematic screening results (section 4).
2. The *reverse cross-attention* mechanism, where descriptor tokens serve as queries attending to encoder features, achieving $163\times$ fewer attention operations than standard cross-attention while delivering competitive performance (section 3.5).
3. Causal analysis demonstrating that the audio-side descriptor (A4) drives the retrieval gains, while the MIDI-side descriptor (D4) acts mainly as a training regularizer and has only weak inference-time effect in top dual models (section 5.2).
4. CKA analysis revealing that descriptors double cross-encoder representational alignment, suggesting the mechanism is representational convergence rather than feature concatenation (section 5.4).

2 Related Work

2.1 Audio–MIDI Matching

Classical audio–MIDI alignment relies on intermediate representations that abstract away modality-specific details. Chromagram-based methods [Ellis, 2007, Müller, 2015] project both modalities into a shared pitch-class energy space, then apply DTW [Hu et al., 2003] or subsequence matching. Audio fingerprinting approaches, pioneered by Shazam [Wang, 2003], extract sparse time-frequency landmarks and match via hash voting, achieving real-time performance but requiring explicit feature engineering. Score-informed source separation [Ewert and Müller, 2012] uses MIDI as a guide for audio decomposition, demonstrating that symbolic information can meaningfully constrain audio processing.

2.2 Self-Supervised Audio Representations

The success of BERT-style pre-training [Devlin et al., 2019] has inspired self-supervised approaches for audio. wav2vec 2.0 [Baevski et al., 2020] learns speech representations via masked prediction over quantized latent features. MERT [Li et al., 2024] extends this paradigm to music, pre-training on 160K hours of music audio and achieving strong performance on downstream MIR tasks. CLAP [Wu et al., 2023] and MuLan [Huang et al., 2022] learn joint audio-language spaces via contrastive objectives, demonstrating that cross-modal alignment can emerge from large-scale training.

2.3 Cross-Modal Contrastive Learning

VICReg [Bardes et al., 2022] regularizes representation learning by jointly optimizing *variance* (preventing collapse), *invariance* (aligning matched pairs), and *covariance* (decorrelating dimensions). Unlike InfoNCE-based methods [He et al., 2020, Chen et al., 2020], VICReg does not require negative pairs or large batch sizes, making it suitable for settings with limited data diversity. Barlow Twins [Zbontar et al., 2021] achieves similar goals by minimizing cross-correlation redundancy. CLIP [Radford et al., 2021] scales contrastive learning to 400M image-text pairs, demonstrating that cross-modal alignment can produce powerful zero-shot representations.

2.4 Feature Conditioning in Neural Networks

Feature-wise Linear Modulation (FiLM) [Perez et al., 2018] conditions network activations via learned affine transformations $\gamma \odot F + \beta$, originally proposed for visual reasoning. Mixture-of-Experts (MoE) [Shazeer et al., 2017] routes inputs through specialized sub-networks via learned gating. Cross-attention [Vaswani et al., 2017, Jaegle et al., 2021] allows one set of representations to attend to another, enabling flexible information flow between modalities or feature streams.

2.5 Representational Similarity Analysis

Centered Kernel Alignment (CKA) [Kornblith et al., 2019] provides a principled measure of representational similarity between neural network layers, invariant to orthogonal transformations and isotropic scaling. Representational Similarity Analysis (RSA) [Kriegeskorte et al., 2008] compares representations via pairwise distance correlations. Linear probing [Alain and Bengio, 2017] evaluates the linear decodability of specific features from frozen representations, providing a complementary view to geometric similarity measures.

3 Method

3.1 Problem Formulation

Given an audio segment $a \in \mathbb{R}^{T_a}$ sampled at $f_s = 24\,\text{kHz}$ ($T_a = 96{,}000$ for 4-second segments) and a corresponding MIDI segment $m = \{(p_i, v_i, d_i, o_i)\}_{i=1}^{N}$ where $p_i \in \{0, \ldots, 127\}$ is pitch, $v_i \in \{0, \ldots, 127\}$ is velocity, d_i is duration, and o_i is onset time, we seek to learn encoders $f_a : \mathbb{R}^{T_a} \to \mathbb{R}^D$ and $f_m : \mathcal{M} \to \mathbb{R}^D$ mapping both modalities into a shared D-dimensional embedding space (with $D = 256$) such that

$$\text{sim}(f_a(a), f_m(m)) \gg \text{sim}(f_a(a), f_m(m')) \quad \text{for matched } (a, m) \text{ and unmatched } m' \neq m. \tag{1}$$

We define the primary evaluation metric as

$$S = \min\big(\text{R@10}_{a\to m},\ \text{R@10}_{m\to a}\big), \tag{2}$$

where $\text{R@10}_{a\to m}$ is Recall@10 for audio-to-MIDI retrieval and $\text{R@10}_{m\to a}$ is the reverse. The $\min$ ensures that both retrieval directions are simultaneously effective.

3.2 Base Architecture

Our architecture consists of modality-specific encoders followed by projection heads that map to the shared embedding space (fig. 1). Full encoder specifications and parameter breakdowns are provided in section B.

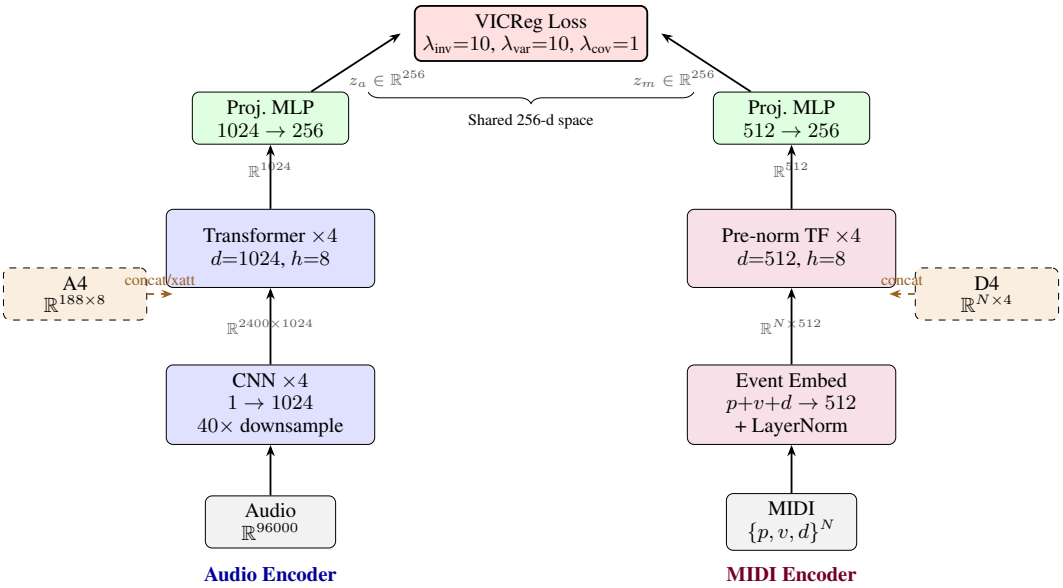

Figure 1: Model architecture for the canonical audio–MIDI baseline. The audio branch uses a 4-layer CNN plus a 4-layer Transformer; the MIDI branch uses event embeddings plus a 4-layer pre-norm Transformer. Both project to a shared 256-D space and train with VICReg. Dashed arrows mark optional descriptor injection points (A4 on audio, D4 on MIDI).

Audio encoder (MERTEncoderLite). The audio encoder follows the MERTEncoderLite architecture [Li et al., 2024]: a 4-layer 1D CNN with GroupNorm and GELU activations ($1 \to 512 \to 512 \to 512 \to 1024$ channels, $40\times$ total downsampling) produces $h \in \mathbb{R}^{B\times 2400\times 1024}$ frame features, which are summed with learnable positional embeddings and processed by a 4-layer Transformer encoder (d_{model}=1024, n_{heads}=8, d_{ff}=4096, GELU, dropout 0.1). The final representation $z_a \in \mathbb{R}^{B\times 1024}$ is obtained via mean pooling over the Transformer output.

MIDI encoder. The MIDI encoder processes discrete event sequences through learned embeddings and a pre-norm Transformer. Each note event's pitch ($\mathbb{R}^{256}$), velocity ($\mathbb{R}^{128}$), and duration ($\mathbb{R}^{128}$) are independently embedded, concatenated to form a 512-dimensional vector, linearly projected, and LayerNorm-ed. Sinusoidal positional encodings are added, and the sequence is processed by a 4-layer pre-norm Transformer (d_{model}=512, n_{heads}=8, d_{ff}=2048, GELU, dropout 0.1). The output $z_m \in \mathbb{R}^{B\times 512}$ is obtained via masked mean pooling over valid (non-padding) positions, followed by a final LayerNorm.

Projection heads. Both encoders are followed by 3-layer MLP projection heads with BatchNorm and ReLU after each hidden layer, mapping to the shared 256-dimensional space. The audio head has dimensions $1024 \to 512 \to 512 \to 256$; the MIDI head is $512 \to 512 \to 512 \to 256$.

Parameter counts. The base model (D0, no descriptors) contains approximately 74.2M parameters: $\sim$60M in the audio encoder, $\sim$13M in the MIDI encoder, and $\sim$1.2M in both projection heads.

3.3 VICReg Loss

We train with VICReg [Bardes et al., 2022], which jointly optimizes three terms over a batch of B matched pairs $\{(z_a^{(i)}, z_m^{(i)})\}_{i=1}^{B}$ projected to embeddings $\hat{z}_a = g_a(z_a)$ and $\hat{z}_m = g_m(z_m)$:

$$\mathcal{L}_{\text{VICReg}} = \lambda_{\text{inv}}\,\mathcal{L}_{\text{inv}} + \lambda_{\text{var}}\,\mathcal{L}_{\text{var}} + \lambda_{\text{cov}}\,\mathcal{L}_{\text{cov}}, \tag{3}$$

The invariance term $\mathcal{L}_{\text{inv}}$ minimizes the MSE between matched embedding pairs; the variance term $\mathcal{L}_{\text{var}}$ applies a hinge loss on the per-dimension standard deviation to prevent representational collapse; and the covariance term $\mathcal{L}_{\text{cov}}$ penalizes off-diagonal elements of the embedding covariance matrix to

Table 1: A4 octave-band definitions. STFT bins correspond to $n_{\text{FFT}} = 2048$ at $f_s = 24\,\text{kHz}$.

Band	Frequency range	STFT bins	Width (Hz)	Musical content
0	47 – 94 Hz	4 – 8	47	Bass fundamentals
1	94 – 188 Hz	8 – 16	94	Bass harmonics
2	188 – 375 Hz	16 – 32	187	Low-mid register
3	375 – 750 Hz	32 – 64	375	Mid register
4	750 – 1500 Hz	64 – 128	750	Upper-mid/presence
5	1500 – 3000 Hz	128 – 256	1500	Presence/harmonics
6	3000 – 6000 Hz	256 – 512	3000	Brilliance
7	6000 – 12000 Hz	512 – 1025	6000	Air/high harmonics

decorrelate dimensions [see Bardes et al., 2022, for full definitions]. We use $\lambda_{\text{inv}} = 10$, $\lambda_{\text{var}} = 10$, $\lambda_{\text{cov}} = 1$ throughout all experiments.

3.4 Descriptors

We design two domain-specific descriptors for the modalities represented in Gate 5B. Additional descriptors explored during screening are detailed in section A.

3.4.1 A4: Octave-Band Energy Dynamics (Audio)

The A4 descriptor captures temporal dynamics of spectral energy across logarithmically-spaced frequency bands. Given an audio waveform, we compute:

1. **STFT**: Compute the magnitude spectrogram with $n_{\text{FFT}} = 2048$, hop length $= 512$, yielding frequency resolution $\Delta f \approx 11.72\,\text{Hz/bin}$ at $f_s = 24\,\text{kHz}$. The output has $T_{\text{stft}} = 188$ frames for a 4-second segment.
2. **Log-magnitude**: $M[k,t] = \log(1 + |X[k,t]|)$, where X is the STFT.
3. **Octave-band aggregation**: We define 8 bands with logarithmically-spaced boundaries (table 1), and average the log-magnitude within each band:
$$\bar{M}[b,t] = \frac{1}{k_b^{\text{end}} - k_b^{\text{start}}} \sum_{k=k_b^{\text{start}}}^{k_b^{\text{end}}-1} M[k,t]. \tag{4}$$
4. **Temporal delta**: $\Delta M[b,t] = \bar{M}[b,t+1] - \bar{M}[b,t]$.
5. **Normalization**: Per-band z-score normalization within each sample:
$$A4[b,t] = \frac{\Delta M[b,t] - \mu_b}{\sigma_b + \epsilon}, \quad \mu_b = \frac{1}{T}\sum_t \Delta M[b,t], \quad \sigma_b = \sqrt{\frac{1}{T}\sum_t (\Delta M[b,t] - \mu_b)^2}. \tag{5}$$

The resulting descriptor $A4 \in \mathbb{R}^{B\times 188\times 8}$ captures how spectral energy *changes* over time in each frequency band, providing the audio encoder with explicit temporal dynamics information.

3.4.2 D4: Local Pitch Intervals (MIDI)

The D4 descriptor encodes local melodic context for each MIDI note using 4 features derived from pitch differences with neighboring notes:

$$D4_i = \begin{bmatrix} (p_i - p_{i-1})/24 \\ (p_{i+1} - p_i)/24 \\ \text{clamp}\big((p_i - p_{i-1})/12,\ -2,\ 2\big)/2 \\ \text{clamp}\big((p_{i+1} - p_i)/12,\ -2,\ 2\big)/2 \end{bmatrix} \in \mathbb{R}^4, \tag{6}$$

where p_i is the MIDI pitch (0–127). The first two features capture raw semitone intervals normalized by two octaves, while the latter two apply clamping to emphasize intervals within ± 2 octaves. Boundary notes ($i = 1$ or $i = N$) use zero-padding.

The output $D4 \in \mathbb{R}^{B\times N\times 4}$ provides the MIDI encoder with explicit intervallic context, analogous to how music theory emphasizes relative pitch over absolute pitch.

3.5 Injection Mechanisms

We explore four mechanisms for injecting descriptors into encoders.

3.5.1 Concatenation

The simplest approach concatenates the descriptor with encoder-internal features and projects back to the original dimension:

$$h'_t = \mathrm{LayerNorm}\big(W_{\text{proj}}[h_t \,\|\, d_t]\big), \tag{7}$$

where $h_t \in \mathbb{R}^{d_{\text{model}}}$ is the encoder feature at time t, d_t is the descriptor value, and $W_{\text{proj}} \in \mathbb{R}^{d_{\text{model}} \times (d_{\text{model}} + d_{\text{desc}})}$.

For A4 injection into the audio encoder: $h_t \in \mathbb{R}^{1024}$, $d_t \in \mathbb{R}^{8}$, so $W_{\text{proj}} \in \mathbb{R}^{1024 \times 1032}$. The concatenation occurs after the CNN and before the Transformer.

For D4 injection into the MIDI encoder: $e_t \in \mathbb{R}^{512}$, $d_t \in \mathbb{R}^{4}$, so $W_{\text{proj}} \in \mathbb{R}^{512 \times 516}$. The concatenation occurs after event embedding and before the Transformer.

3.5.2 Standard Cross-Attention

In standard cross-attention, encoder features serve as queries attending to descriptor features:

$$\mathrm{XAttn}(F, D) = \mathrm{softmax}\left(\frac{Q_F\, K_D^\top}{\sqrt{d_k}}\right) V_D, \tag{8}$$

where $Q_F = F\, W_Q \in \mathbb{R}^{T_F \times d_k}$, $K_D = D_{\text{proj}}\, W_K \in \mathbb{R}^{T_D \times d_k}$, $V_D = D_{\text{proj}}\, W_V \in \mathbb{R}^{T_D \times d_v}$. Here $T_F = 2400$ (CNN output frames) and T_D depends on the descriptor.

The attention complexity is $O(T_F \cdot T_D \cdot d_k)$, dominated by the $T_F = 2400$ query tokens.

3.5.3 Reverse Cross-Attention (Key Contribution)

We propose *reverse cross-attention*, where **descriptor tokens serve as queries** attending to encoder features:

$$\mathrm{RevXAttn}(D, F) = \mathrm{softmax}\left(\frac{Q_D\, K_F^\top}{\sqrt{d_k}}\right) V_F, \tag{9}$$

where $Q_D = D_{\text{proj}}\, W_Q \in \mathbb{R}^{T_D \times d_k}$, $K_F = F\, W_K \in \mathbb{R}^{T_F \times d_k}$, $V_F = F\, W_V \in \mathbb{R}^{T_F \times d_v}$.

For A4 reverse cross-attention: $T_D = 188$ (native STFT frames) and $T_F = 2400$ (CNN output). The descriptor is projected to $d_{\text{model}} = 1024$ and augmented with learnable positional embeddings before serving as queries.

Complexity advantage. The critical difference is in the *output* sequence length: reverse cross-attention produces $T_D = 188$ tokens (which are then mean-pooled), while standard cross-attention produces $T_F = 2400$ tokens. Since subsequent Transformer self-attention has $O(T^2)$ complexity, processing 188 vs. 2400 tokens yields:

$$\text{Speedup} = \frac{T_F^2}{T_D^2} = \frac{2400^2}{188^2} \approx 163\,\times\,. \tag{10}$$

This is conceptually related to the Perceiver architecture [Jaegle et al., 2021], which uses a small set of latent tokens as queries to compress high-dimensional inputs. In our case, the "latent" tokens are not learned from scratch but carry structured domain knowledge via the descriptor.

3.5.4 Combined Mechanisms

Two combined configurations are evaluated in Gate 5B:

- **d4a4**: D4 concatenation in MIDI encoder + A4 concatenation in audio encoder.
- **d4-a4r**: D4 concatenation in MIDI encoder + A4 reverse cross-attention in audio encoder.

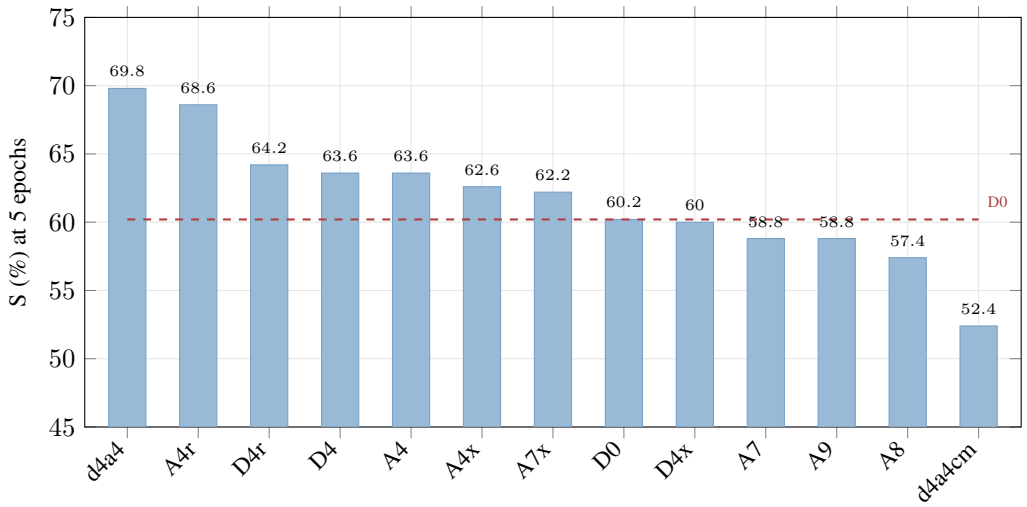

Figure 2: Gate 4.3 screening results: 13 descriptor×mechanism combinations ranked by S at 5 training epochs. Dashed line marks D0 baseline (S=60.2%). The top 7 arms all inject descriptors into their own modality's encoder. Cross-modal injection (d4a4cm) is worst.

3.6 Training Protocol

Dataset. We use MAESTRO v3.0.0 [Hawthorne et al., 2019], containing 1,276 classical piano performances with aligned audio and MIDI (∼200 hours). Audio–MIDI pairs are segmented into 4-second windows with 1-second hop, yielding ∼13,000 training segments at $f_s = 24$ kHz.

Optimizer. AdamW [Loshchilov and Hutter, 2019] with weight decay 0.01, $\beta = (0.9, 0.999)$, and batch size 16.

Learning rate schedule. We employ a cosine-with-tail schedule [Loshchilov and Hutter, 2017]: linear warmup for 500 steps, cosine annealing with reference period of 30 epochs, a floor of 10% of base LR, and a linear tail decaying to 2% in the final epoch.

Hardware. Primary experiments on a single NVIDIA RTX 3090 (24 GB). Multi-seed validation and parameter-matched controls on a cluster of 36× NVIDIA A30 (24 GB each) via SLURM. Full hyperparameters and scheduler details are provided in section D.

4 Descriptor and Mechanism Selection

4.1 Short-Horizon Screening (5 Epochs, 13 Arms)

We conduct a systematic screening of 13 descriptor×mechanism combinations over 5 training epochs. Each arm uses identical hyperparameters except for the descriptor and injection mechanism. Results are shown in table 2 and fig. 2.

Key observations.

1. **Same-modality injection dominates**: The top 7 arms all inject descriptors into their own modality's encoder. The single cross-modal injection (d4a4cm, injecting audio features into MIDI and vice versa) is the worst arm (−7.8 pp).
2. **Dual injection is best**: d4a4 (+9.6 pp) combines D4 in MIDI and A4 in audio, exploiting both modalities.
3. **Reverse cross-attention is competitive**: A4r (+8.4 pp) is the second-best arm despite processing only 188 tokens vs. 2400.

Table 2: Gate 4.3 screening results (5 epochs). S = min(R@$10_{a\to m}$, R@$10_{m\to a}$). Δ is relative to the D0 baseline. Arms above the horizontal line outperform D0.

Rank	Arm	Descriptor	Mechanism	S (%)	Hard neg (%)	Δ (pp)
1	**d4a4**	D4+A4	Dual concat	**69.8**	91.6	$+9.6$
2	A4r	A4	Reverse cross-att	68.6	91.6	$+8.4$
3	D4r	D4	Reverse cross-att	64.2	93.2	$+4.0$
4	D4	D4	Concat	63.6	91.2	$+3.4$
4	A4	A4	Concat	63.6	92.4	$+3.4$
6	A4x	A4	Cross-attention	62.6	92.4	$+2.4$
7	A7x	A7	Cross-attention	62.2	92.0	$+2.0$
8	**D0**	—	*Baseline*	**60.2**	90.0	—
9	D4x	D4	Cross-attention	60.0	91.4	-0.2
10	A7	A7	Concat	58.8	90.2	-1.4
10	A9	A9	Concat	58.8	90.4	-1.4
12	A8	A8	Concat	57.4	90.6	-2.8
13	d4a4cm	D4+A4	Cross-modal	52.4	89.6	-7.8

Table 3: Gate 4.4 architecture family screening (5 epochs, from d4a4 foundation). Only Third Tower with weighted loss outperforms the D0 baseline substantially.

Rank	Arm	Family	S (%)	Δ vs D0 (pp)
1	t3-wt	Third Tower	67.6	$+7.4$
2	t3-tri	Third Tower	65.0	$+4.8$
3	moe-a4-v2	MoE	60.2	± 0.0
4	film-dual	FiLM	59.4	-0.8
4	moe-a4-v4	MoE	59.4	-0.8
6	film-a4	FiLM	59.2	-1.0
6	moe-dual	MoE	59.2	-1.0
6	moe-a4-v3	MoE	59.2	-1.0
9	film-d4	FiLM	58.6	-1.6
10	moe-a4	MoE	58.2	-2.0
11	t3-anc	Third Tower	42.2	-18.0

4. **Complex descriptors underperform**: A7 (rational attractor, 12 dim), A8 (onset-chroma, 12 dim), and A9 (IDF-attractor, 12 dim) all perform at or below baseline, while the simpler A4 (8 dim) and D4 (4 dim) excel.

4.2 Architecture Families (Gate 4.4, 11 Arms)

We additionally explore three alternative architectural families for descriptor injection (table 3).

FiLM (γ, β modulation of Transformer features): No variant improves over the baseline, suggesting that affine transformation is too constrained for the cross-modal information flow required.

Mixture-of-Experts (learned routing across 4–8 experts): All variants perform near or below D0, indicating that the routing mechanism does not discover useful specialization patterns for this task.

Third Tower (separate descriptor encoder with auxiliary VICReg loss): The weighted-loss variant (t3-wt) achieves $S = 67.6\%$ ($+7.4$ pp), competitive with direct injection, but the anchor-based variant (t3-anc) collapses (-18.0 pp). Per-family Gate 4.4 results are reported in section C.

4.3 Long-Horizon Confirmation (30–60 Epochs)

The top candidates from screening are trained for 30–60 epochs with their selected schedules (Cosine-30/60, Cosine-tail, Trapezoidal hold). Results (table 4) confirm the screening rankings in the

Table 4: Long-horizon results (30–60 epochs) from the seed-42 training run. Δ: improvement over D0 at convergence. Multi-seed training replication is summarized in table 12.

Arm	Schedule	Epochs	Best S (%)	Best epoch	Params (M)	Δ (pp)
d4a4	Cosine-60	60	**83.8**	50	75.5	$+10.4$
a4r	Cosine-30	30	82.0	29	78.6	$+8.6$
t3-wt	Trapez-50	50	81.2	50	—	$+7.8$
a4r	Cosine-tail	60	80.6	60	78.6	$+7.2$
d4-a4r	Cosine-30	30	79.8	30	78.9	$+6.4$
D0	Cosine-tail	60	73.4	50	74.2	—

Table 5: Canonical evaluation results (Test 12) from the seed-42 checkpoint. Pool size 256, 500 queries, seed 42. Multi-seed training replication is summarized in table 12.

Arm	S (%)	R@1$_{a\to m}$	R@10$_{a\to m}$	MRR$_{a\to m}$	R@10$_{m\to a}$	MRR$_{m\to a}$	Hard neg (%)
D0	73.4	22.2	74.8	0.380	73.4	0.372	98.4
d4a4	**83.8**	**28.0**	**84.4**	**0.458**	**83.8**	**0.442**	**99.0**
a4r	82.0	27.6	82.6	0.438	82.0	0.424	98.8
d4-a4r	79.8	25.2	81.4	0.426	79.8	0.420	99.4

canonical seed-42 run, with one notable result: d4a4 reaches $S = 83.8\%$ at epoch 50 under 60-epoch cosine-stretched scheduling.

5 Scientific Validation — Gate 5B

We validate the top 4 models (D0, d4a4, a4r, d4-a4r) through a battery of 8 complementary tests, each probing a different aspect of the learned representations. Unless otherwise noted, the detailed diagnostic tests below are computed on the canonical seed-42 checkpoint. The final subsection reports the five-seed training replication used for the paper's headline claims.

5.1 Canonical Evaluation (Test 12: Scoreboard)

The canonical evaluation uses a structured pool protocol: 256 pool items, 500 queries, 64 hard negatives (same piece, different time), 32 semi-hard negatives (same composer, different piece), seed 42. Results are shown in table 5.

All descriptor-augmented models substantially outperform the D0 baseline: d4a4 by $+10.4$ pp, a4r by $+8.6$ pp, and d4-a4r by $+6.4$ pp. The improvements are consistent across all metrics (R@1, R@10, MRR).

5.2 Causal Ablation (Test 01)

To determine whether descriptors are *causally* necessary at inference time (as opposed to merely useful during training), we ablate each descriptor independently using three perturbation modes: **zero** (replace descriptor with zeros), **noise** (replace with Gaussian noise matching descriptor statistics), and **shuffle** (randomly permute descriptors across the batch).

Results (table 6, fig. 3) reveal a stark asymmetry:

A4 is completely causal. Zeroing A4 collapses S from $83.8\% \to 7.8\%$ (d4a4), $82.0\% \to 4.4\%$ (a4r), and $79.8\% \to 4.4\%$ (d4-a4r)—drops of 75–78 pp that bring performance below random chance. Even the milder shuffle ablation causes drops of 32–37 pp. The model has become *entirely dependent* on A4 for cross-modal alignment.

D4 has weak inference effect in this family. In the dual top models, zeroing D4 in d4a4 produces $S = 84.4\%$ (actually $+0.6$ pp higher than normal), and in d4-a4r produces $S = 79.4\%$ (-0.4 pp,

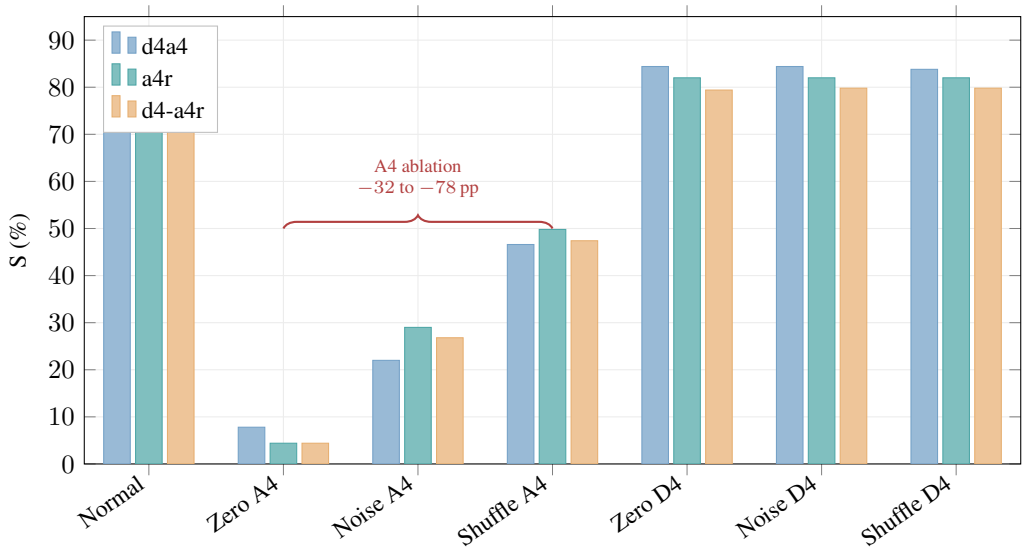

Figure 3: Causal ablation results (Test 01) for the canonical seed-42 checkpoint. A4 ablation (left group) collapses S by 32–78 pp across all models. D4 ablation (right group) has near-zero effect in dual models (-0.4 to $+0.6$ pp). Note: a4r has no D4 descriptor, so its D4 bars equal its Normal bar for visual alignment.

Table 6: Causal ablation results (Test 01). Δ: change in S relative to normal inference. **A4 is causally essential; D4 has weak effect in this model family.**

	Ablation mode	d4a4 (Δ)	a4r (Δ)	d4-a4r (Δ)
Audio (A4)	Zero	-76.0 pp	-77.6 pp	-75.4 pp
	Noise	-61.8 pp	-53.0 pp	-53.0 pp
	Shuffle	-37.2 pp	-32.2 pp	-32.4 pp
MIDI (D4)	Zero	$+0.6$ pp	—	-0.4 pp
	Noise	$+0.6$ pp	—	± 0.0 pp
	Shuffle	± 0.0 pp	—	± 0.0 pp

within evaluation noise). In a D4-only control arm, ablating D4 yields small but non-zero drops (63.6% $\rightarrow$ 62.8% or 62.4%, i.e., 0.8–1.2 pp).

The D4 Paradox. D4 improves training dynamics ($+3.4$ pp over D0 at 5 epochs, and d4a4 outperforms A4-only by consistent margins) but shows weak or near-zero inference-time effect once training converges in top dual models. We hypothesize D4 acts primarily as a *training regularizer*: by providing the MIDI encoder with explicit interval context early in training, it accelerates the discovery of useful representations that the encoder eventually internalizes.

5.3 Transposition Invariance (Test 04)

A musically meaningful test of cross-modal representations is their robustness to pitch transposition: shifting all MIDI pitches by $\pm k$ semitones should preserve the musical content (modulo range effects) while the audio remains unchanged. We evaluate S at transpositions $k \in \{-6, -3, -1, 0, +1, +3, +6\}$ semitones (table 7, fig. 4).

Key finding. The reverse cross-attention models (a4r, d4-a4r) show the highest transposition invariance: at ± 3 semitones, a4r retains 59% of its baseline S compared to 37% for D0—an advantage of $+23.6$ pp in absolute S. This suggests that reverse cross-attention, by forcing the model to attend to encoder features *from the perspective of spectral dynamics*, encourages learning of interval-relative representations that generalize across pitch shifts.

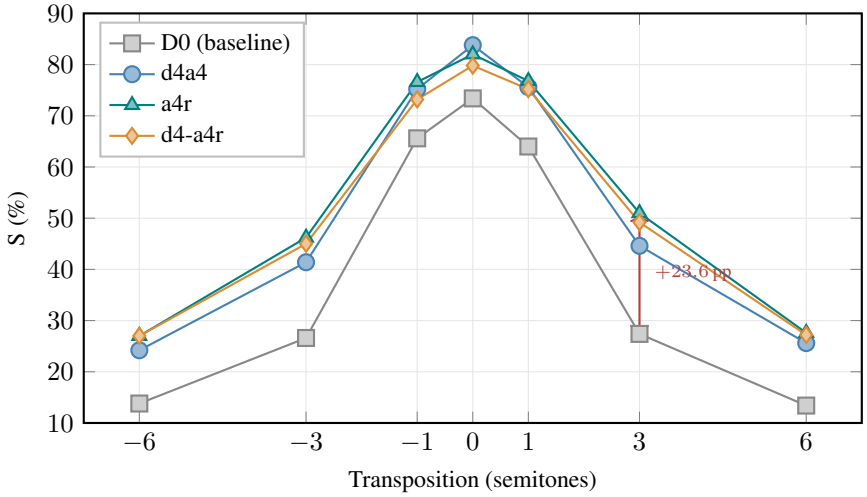

Figure 4: Transposition invariance (Test 04) for the canonical seed-42 checkpoint. All models degrade with pitch shift, but descriptor-augmented models retain more performance. At ± 3 semitones, a4r shows $+23.6$ pp advantage over D0. The reverse cross-attention models (a4r, d4-a4r) are the most invariant.

Table 7: Transposition invariance (Test 04) for the seed-42 checkpoint. S (%) at various semitone shifts. Retention = $S_{|\pm 3|}$ / S_0. Multi-seed training replication is summarized in table 12.

Arm	-6	-3	-1	0	$+1$	$+3$	$+6$	Ret. $\lvert \pm 3 \rvert$
D0	13.8	26.6	65.6	73.4	64.0	27.4	13.4	37%
d4a4	24.2	41.4	75.2	83.8	75.6	44.6	25.6	51%
a4r	**27.0**	**46.2**	**76.6**	82.0	**76.8**	**51.0**	**27.6**	**59%**
d4-a4r	27.0	45.0	73.2	79.8	75.2	49.2	27.2	59%

5.4 Cross-Encoder Alignment — CKA Analysis (Test 06)

To understand *how* descriptors improve performance, we measure the representational similarity between audio and MIDI transformer layers using Centered Kernel Alignment [Kornblith et al., 2019]:

$$\mathrm{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F}, \tag{11}$$

where $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$ are centered activation matrices for n samples across p and q neurons respectively. CKA = 1 indicates identical representations (up to rotation and scaling); CKA = 0 indicates orthogonal representations.

We compute CKA for all 16 audio$\times$MIDI layer pairs (4 audio layers $\times$ 4 MIDI layers) using 500 matched pairs (table 8, fig. 5).

Descriptors double cross-encoder alignment. The mean CKA increases monotonically with descriptor complexity: D0 (0.435) $\rightarrow$ d4a4 (0.659) $\rightarrow$ a4r (0.766) $\rightarrow$ d4-a4r (0.794). Descriptors cause the audio and MIDI transformers to develop *similar internal representations*—they "speak the same language."

Alignment $\neq$ retrieval. Critically, d4-a4r has the highest CKA (0.794) but *not* the highest S (79.8% vs. 83.8% for d4a4). This dissociation suggests that while representational alignment is necessary for good cross-modal retrieval, *excessive* alignment may reduce the capacity for modality-specific information that aids discrimination. The relationship between CKA and S appears to be non-monotonic.

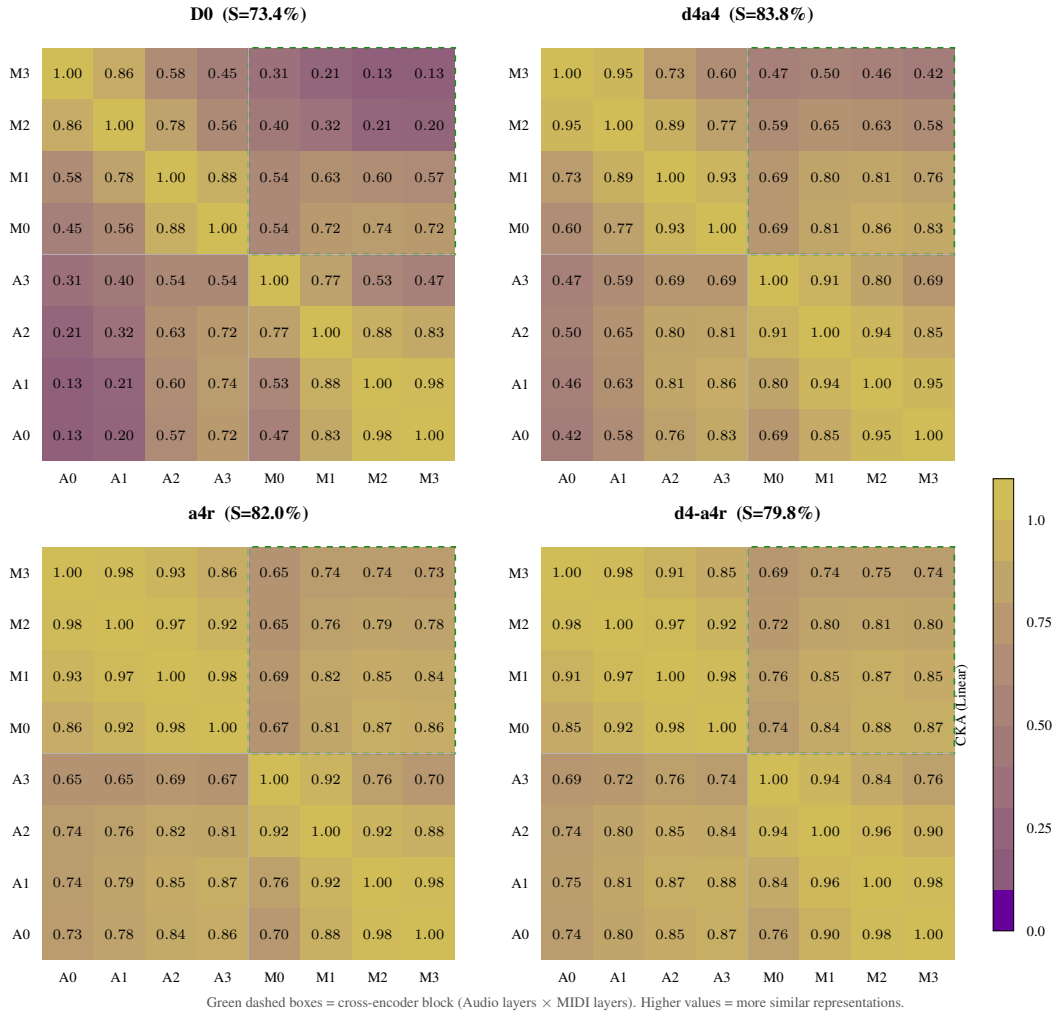

Figure 5: CKA representational alignment across all 8×8 layer pairs (Test 06) for the canonical seed-42 checkpoint. Rows/columns: audio layers (A0–A3) and MIDI layers (M0–M3). Green dashed boxes highlight the cross-encoder block. Within-encoder alignment is uniformly high (> 0.85); cross-encoder alignment reveals the descriptor effect: D0 mean $= 0.435$; d4a4 $= 0.659$ ($+51\%$); a4r $= 0.766$ ($+76\%$); d4-a4r $= 0.794$ ($+82\%$). Alignment increases monotonically with layer depth in all models.

Layer-depth analysis. CKA increases near-monotonically with layer depth in all models (e.g., D0: 0.305 at layer 0×0 to 0.722 at layer 3×3; d4-a4r: 0.686 to 0.873), consistent with the expectation that deeper layers develop more abstract, modality-invariant representations.

5.5 Spectral Band Sensitivity (Test 08)

We probe which A4 octave bands carry the most discriminative information by applying perturbations ($\epsilon = 0.1$) to individual bands and measuring the mean L_2 change in the output embedding (table 9, fig. 6).

High-frequency dominance. In all models, bands 4–7 (750–12000 Hz) carry 2–10× more discriminative power than bands 0–3. This is musically meaningful: the upper harmonic series (> 750 Hz) carries the most information about harmonic relationships and timbral identity, while low-frequency fundamentals are more uniform across pieces.

Table 8: Cross-encoder CKA alignment (Test 06). Mean CKA across all 16 audio×MIDI layer pairs. Higher = more aligned internal representations between encoders.

Arm	CKA mean	Δ vs D0	RSA mean
D0	0.435	—	0.446
d4a4	0.659	+51%	0.646
a4r	0.766	+76%	0.721
d4-a4r	**0.794**	**+82%**	**0.761**

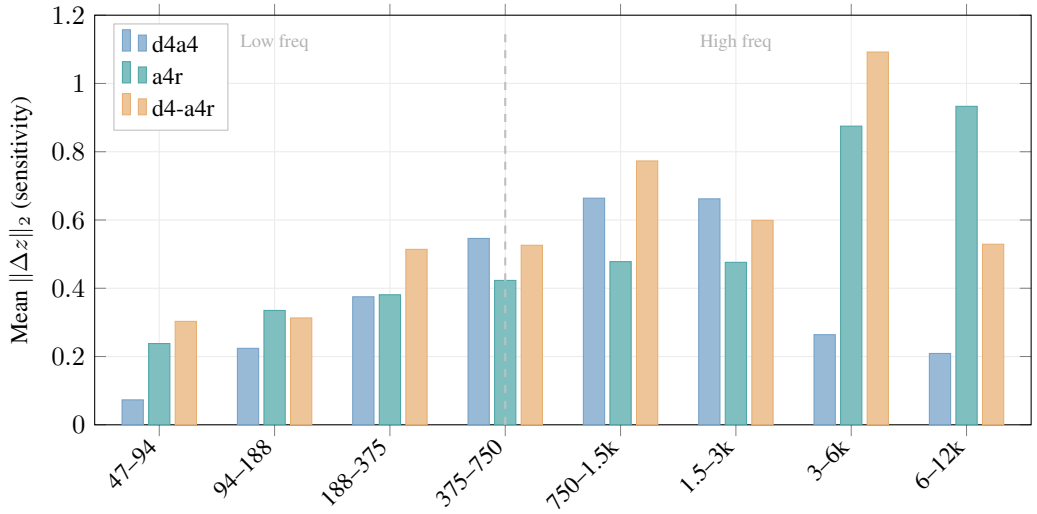

Figure 6: Perturbation sensitivity by A4 octave band (Test 08) for the canonical seed-42 checkpoint. X-axis labels show full band ranges in kHz/Hz. Each group shows the mean L_2 change in output embedding when perturbing one band by ϵ=0.1. High-frequency bands (750+ Hz) carry 2–10× more sensitivity. Each injection mechanism attends to different spectral regions: d4a4 peaks at 750–3000 Hz, a4r at 3–12 kHz, d4-a4r at 3–6 kHz.

Mechanism-dependent spectral focus. Each injection mechanism "listens" to different spectral regions: d4a4 (concatenation) is most sensitive to bands 4–5 (750–3000 Hz), while a4r (reverse cross-attention) is most sensitive to bands 6–7 (3000–12000 Hz). The d4-a4r combination peaks at band 6 (3000–6000 Hz, sensitivity 1.092).

Non-linear encoding. Despite high perturbation sensitivity, the Pearson correlation between A4 band values and output embedding dimensions is uniformly low ($|r| < 0.05$), indicating that the descriptor information is encoded through *highly non-linear transformations* rather than being linearly preserved.

5.6 Input Invariance (Test 09)

We assess robustness to four input perturbations: temporal misalignment, MIDI velocity scaling, octave transposition, and additive audio noise (table 10 and fig. 7).

Temporal robustness. At ±8000 samples (±0.33 s at 24 kHz), worst-case drops range from −3.6 pp (d4-a4r: 79.8 → 76.2) to −7.2 pp (d4a4: 83.8 → 76.6). D0 drops −5.2 pp (73.4 → 68.2). All models retain $> 90\%$ of baseline S, which is expected given that a 0.33 s shift is ∼8% of the 4-second segment duration.

Velocity fragility. All models collapse at 0.5× ($S = 5$–9%) and 1.5× ($S = 12$–18%). Even 0.8× causes large drops. The degradation is asymmetric: S remains at 44–55% absolute at 1.2× vs. only 32–47% at 0.8×. Note that velocity scaling modifies only MIDI velocity values, not note durations

Table 9: Perturbation sensitivity by A4 octave band (Test 08). Values are mean $\|\Delta z\|_2$ when perturbing each band by $\epsilon = 0.1$. Higher = more sensitive. Bands 4–7 (750+ Hz) dominate.

Band	Hz range	d4a4	a4r	d4-a4r	Zone
0	47 – 94	0.073	0.238	0.303	Low
1	94 – 188	0.224	0.335	0.313	Low
2	188 – 375	0.375	0.381	0.514	Mid
3	375 – 750	0.546	0.423	0.526	Mid
4	750 – 1500	**0.664**	0.478	**0.773**	High
5	1500 – 3000	**0.662**	0.476	0.599	High
6	3000 – 6000	0.264	**0.875**	**1.092**	High
7	6000 – 12000	0.209	**0.933**	0.529	High

Table 10: Perturbation robustness (Test 09). Each cell shows worst-case S (%) across perturbation directions at the given level. Temporal shift values are in samples (24 kHz sample rate).

Perturbation	Level	D0	d4a4	a4r	d4-a4r
Temporal shift	$\pm$8000 samp ($\pm$0.33 s)	68.2	76.6	75.0	76.2
Velocity scale	0.5$\times$	5.2	8.8	6.8	6.4
Velocity scale	1.5$\times$	18.4	12.8	12.4	11.6
Octave transp.	$-$12 st	12.0	16.0	17.4	17.0
Octave transp.	$+$12 st	10.0	13.8	16.0	15.8

or onsets. The MIDI encoder learns a specific velocity→embedding mapping during training; scaling distorts this mapping because the model has never seen velocity values outside the natural training distribution. The asymmetry may reflect two effects: at 0.5$\times$, many notes are pushed toward velocity ≈ 0, collapsing dynamic range; at 1.5$\times$, values are clamped to 127, saturating the upper range. Velocity augmentation during training is a natural candidate for improving robustness.

Octave catastrophe. All models collapse at $\pm$12 semitones ($S = 10$–17%). The a4r and d4-a4r models retain slightly higher S (16–17%) than D0 and d4a4 (10–16%), consistent with the transposition advantage seen in Test 04 at smaller intervals. Combined with Test 04 (section 5.3), this confirms that all models learn pitch-absolute representations; the partial invariance at $\pm$1–3 semitones does not extend to octave-scale shifts.

Audio noise: non-monotonic ranking. D0 is most robust at moderate noise (20 dB: -0.4 pp vs. -29 pp for d4a4), because it has no A4 descriptor pathway that can be corrupted. However, at 5 dB the ranking reverses: d4-a4r (33.0%) $>$ a4r (31.8%) $>$ d4a4 (25.0%) $>$ D0 (17.8%). At moderate noise, descriptor corruption dominates (A4 is computed from noisy audio). At extreme noise, the MIDI encoder—which is noise-invariant by construction—provides a floor that is higher in descriptor-augmented models due to their stronger cross-modal alignment (CKA, Test 06). D0's MIDI encoder lacks this alignment and collapses when the audio side is destroyed. The mechanism-dependent spectral sensitivity (Test 08, section 5.5) may explain why a4r/d4-a4r are more sensitive at moderate noise—they rely more heavily on high-frequency bands (3000–12000 Hz), which are first corrupted by additive Gaussian noise.

5.7 Linear Probing (Test 03)

We evaluate the linear decodability of domain-specific features from frozen embeddings using 6 diagnostic probes (table 11).

Counterintuitive result. D0 (no descriptors) achieves the highest R^2 on the most informative cross-modal probe (MIDI $\rightarrow$ chroma: 0.330 vs. $\sim$0.25 for descriptor models). This suggests that descriptor-augmented models encode cross-modal information in a way that is *less linearly decodable* but more effective for retrieval. The discriminative advantage of descriptors lives in distance geometry (cosine similarity structure) rather than in linearly accessible features.

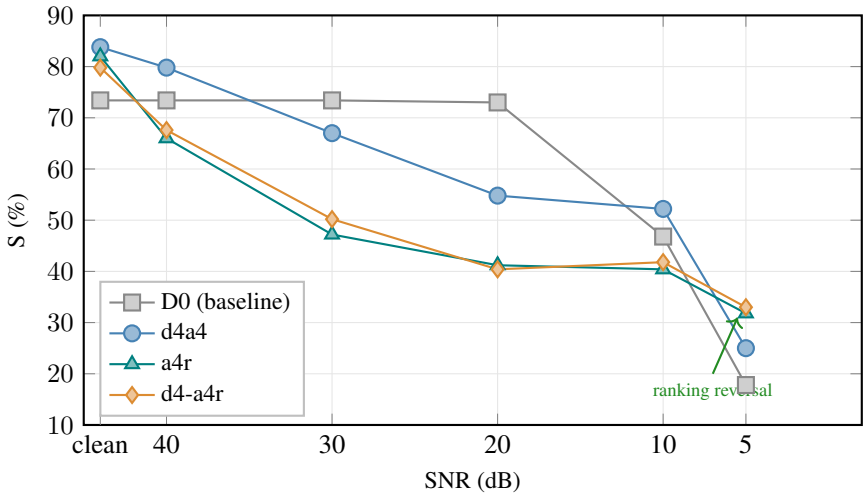

Figure 7: Audio noise degradation (Test 09) for the canonical seed-42 checkpoint. At moderate noise ($\geq$ 20 dB), D0 is most robust because it has no corruptible A4 pathway. At 5 dB the ranking reverses: descriptor models retain higher absolute S, likely due to their stronger cross-modal alignment providing a noise-invariant floor via the MIDI encoder.

Table 11: Linear probe R^2 from frozen 256-dim embeddings (Test 03). Cross-modal probes decode features of one modality from the other modality's embedding.

Type	Probe	D0	d4a4	a4r	d4-a4r
Cross	Audio $\rightarrow$ pitch histogram	0.181	0.174	0.167	0.186
	Audio $\rightarrow$ interval hist.	0.094	0.112	0.095	0.115
	MIDI $\rightarrow$ chroma	**0.330**	0.245	0.255	0.251
	MIDI $\rightarrow$ centroid	0.616	0.637	**0.662**	0.652
Self	Audio $\rightarrow$ chroma	0.310	0.235	0.249	0.231
	MIDI $\rightarrow$ pitch histogram	0.239	0.236	0.233	0.233

5.8 Embedding Space Visualization (Test 10)

We visualize the 256-dimensional embedding spaces using t-SNE [van der Maaten and Hinton, 2008] and UMAP [McInnes et al., 2018] with 2000 samples per model. Audio and MIDI embeddings are plotted together, color-coded by modality, with lines connecting matched pairs. Additionally, we compute the distribution of cosine similarities between matched audio–MIDI pairs (figs. 9 to 11).

The visualizations confirm quantitative findings: descriptor-augmented models show tighter clustering of matched pairs and higher mean cosine alignment (d4a4: 0.72 mean cosine vs. D0: 0.58). A compact summary dashboard of the seed-42 validation results is shown in fig. 8; full t-SNE/UMAP details and cosine-alignment distribution are provided in section E.

5.9 Multi-Seed Training Replication

The headline performance claims in this paper are anchored in a five-seed training replication over the four canonical arms. This summary should be read separately from the seed-42 diagnostic tables above: it reports training variability across independent reruns rather than within-checkpoint structure.

The replicated ordering remains stable across reruns. For the flagship comparison, d4a4 exceeds D0 by $+8.8$ pp on mean S, with a large effect size (Cohen $d = 3.51$; Welch $t = 5.55$; $p < 0.001$). The observed five-seed ranges do not overlap: the worst guided seed (81.4) remains 4.0 percentage points above the best unguided seed (77.4).

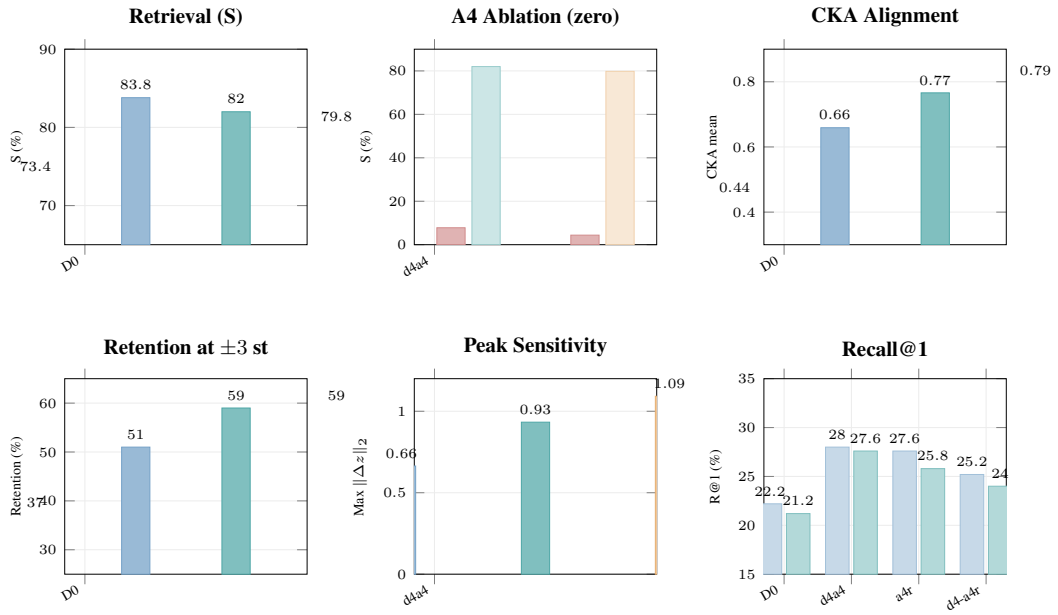

Figure 8: Seed-42 summary dashboard: retrieval S, collapse under A4 ablation, CKA alignment, transposition retention at ± 3 semitones, peak perturbation sensitivity, and bidirectional Recall@1.

Table 12: Multi-seed training replication across the four canonical arms. Means, standard deviations, and ranges are computed over five independent training seeds.

Arm	Mean S (%)	SD (pp)	Range	Δ vs D0 (pp)
d4a4	**84.0**	**2.7**	**81.4–87.6**	**+8.8**
d4-a4r	81.2	2.5	78.4–83.4	+6.0
a4r	80.7	1.9	79.4–84.0	+5.5
D0	75.2	2.3	71.8–77.4	—

6 Discussion

6.1 The A4 Mechanism: Spectral Dynamics as Cross-Modal Bridge

The most striking finding of this study is the complete causal dependence on A4—a relatively simple 8-dimensional descriptor capturing temporal changes in octave-band energy. The model does not merely *use* A4 as supplementary information; it has reorganized its entire representational strategy around it. When A4 is removed, performance collapses to below random chance ($S < 8\%$), far worse than the 73.4% achieved by D0 which never had access to A4.

This suggests that A4 provides a *bridge representation*: a feature that is simultaneously computable from audio (via STFT) and predictable from MIDI (via expected harmonic content). Temporal dynamics of spectral energy—how energy in each octave band changes from frame to frame—capture the rhythmic and harmonic contour of a performance in a way that transcends the audio/symbolic divide.

6.2 The D4 Paradox: Training Regularization Without Inference Causality

D4 consistently improves screening performance ($+3.4$ pp at 5 epochs) and contributes to d4a4 achieving the all-time record, yet in top dual models its ablation effect at inference is near-zero (-0.4 to $+0.6$ pp). In the D4-only control arm, the effect is small but non-zero (0.8–1.2 pp). We propose three complementary explanations:

1. **Curriculum effect**: D4 provides explicit interval context that helps the MIDI encoder discover useful features faster, but the encoder eventually learns equivalent representations from raw note sequences.
2. **Gradient diversity**: D4 features create additional gradient pathways during backpropagation, improving optimization landscape navigation without remaining necessary in the forward pass.
3. **Implicit regularization**: The projection from 516 to 512 dimensions (eq. (7)) acts as a bottleneck that encourages the encoder to compress descriptor information into its internal representations.

6.3 Reverse Cross-Attention: Informational Bottleneck as Feature

In the canonical seed-42 comparison, the reverse cross-attention mechanism (a4r) achieves $S = 82.0\%$—competitive with the 83.8% record of d4a4—while being $163\times$ cheaper in attention operations. Beyond computational efficiency, a4r shows the best transposition invariance (+23.6 pp advantage over D0 at ± 3 semitones). We attribute this to the *informational bottleneck* created by using only 188 descriptor tokens as queries: by forcing all encoder information to flow through spectral-dynamics-shaped queries, the model is encouraged to learn representations structured around how energy changes across frequency bands—a property that is inherently invariant to absolute pitch.

6.4 Alignment and Retrieval Are Not Monotonic

The CKA analysis reveals a counterintuitive dissociation: d4-a4r has the highest cross-encoder alignment (CKA = 0.794) but not the highest retrieval performance ($S = 79.8\%$ vs. 83.8% for d4a4 with CKA = 0.659). This suggests a non-monotonic relationship between representational alignment and task performance. We hypothesize that moderate alignment preserves sufficient modality-specific information for discrimination, while excessive alignment homogenizes representations at the cost of discriminative detail.

6.5 Noise Robustness and Descriptor Vulnerability

Descriptor-augmented models show larger relative S drops than D0 at moderate noise levels (≥ 20 dB SNR), but at 5 dB the ranking reverses and descriptor models retain higher absolute S (fig. 7). The noise robustness profile is thus non-monotonic: A4 acts as both a strength and a vulnerability—it provides a powerful cross-modal bridge but creates a new attack surface through the audio-computed descriptor pathway. At moderate noise, A4 corruption dominates; at extreme noise, the superior cross-modal alignment of descriptor models (CKA 0.66–0.79 vs. 0.44 for D0, section 5.4) provides residual matching ability through the noise-invariant MIDI pathway. Applications in noisy environments should consider the full robustness profile alongside peak performance.

6.6 Limitations

1. **Single dataset**: All experiments use MAESTRO v3.0.0, which contains exclusively classical piano. Generalization to other instruments, genres, or polyphonic settings is unknown.
2. **Headline replication is now closed**: The flagship d4a4 arm has been replicated across five independent training seeds, yielding $84.0\% \pm 2.7$ pp. The observed five-seed ranges do not overlap with D0, and the effect size against the baseline remains large (Cohen $d = 3.51$, $p < 0.001$).
3. **Parameter-matched controls are now closed**: Test 02 isolates descriptor content from parameter-count inflation within the same architectural family. The real arm reaches 83.0%, while the parameter-matched zero, random, and shuffled controls fall to 75.0%, 73.6%, and 73.6%, respectively.
4. **Audio encoder partially frozen**: The MERTEncoderLite architecture is partially initialized from pre-trained weights, potentially confounding the analysis of descriptor effects with pre-training biases.
5. **No input augmentation**: All models are trained on unperturbed data. Velocity scaling and octave transposition cause catastrophic performance drops ($S < 18\%$), suggesting that data augmentation during training could substantially improve robustness (see Test 09, section 5.6).

6.7 Future Directions

1. **Other instruments and genres**: Testing on datasets with broader timbral diversity (e.g., MusicNet [Thickstun et al., 2017]).
2. **Descriptor discovery**: Using the A4 success as a template for systematic descriptor search, potentially via learned descriptors initialized from hand-crafted ones.
3. **Scaling**: Exploring whether the descriptor injection paradigm scales to larger encoders and datasets.
4. **Robustness through augmentation**: Incorporating velocity scaling, octave transposition, and audio corruption during training to test whether the current fragilities can be reduced without erasing the descriptor-guided advantage.

7 Conclusion

We have presented a systematic exploration of descriptor injection for cross-modal audio–MIDI learning, spanning 24 experimental configurations across 3 phases of increasing rigor. Our findings challenge the assumption that cross-modal improvement requires complex architectural innovations: a simple 8-dimensional spectral dynamics descriptor (A4), injected via concatenation, yields a replicated mean advantage of $+8.8$ percentage points over a strong VICReg baseline.

The scientific validation reveals that this improvement is not merely additive feature concatenation. Causal ablation shows the model has become *entirely dependent* on A4, reorganizing its representations around spectral dynamics. CKA analysis reveals that the mechanism is *representational convergence*: descriptors cause the audio and MIDI encoders to develop similar internal representations, effectively teaching both modalities a shared language grounded in spectral dynamics.

The reverse cross-attention mechanism offers both practical ($163\times$ cheaper) and scientific (enhanced transposition invariance) advantages, demonstrating that structured information bottlenecks can improve generalization. The D4 paradox—clear training benefit with weak inference-time causality in this model family—suggests that domain-specific descriptors may play a role as training regularizers even when their forward-pass contribution is limited at deployment.

These results position descriptor injection as a lightweight, effective paradigm for cross-modal learning in the music domain, and suggest that the principle may generalize to other cross-modal settings where domain knowledge can be encoded as structured features. This work is situated within a broader theoretical program—Harmonic Information Theory—which examines the hypothesis that frequency ratios may function as informational carriers across physical modalities. A companion volume develops the full theoretical framework [Fernández Méndez and Echániz, 2026].

AI assistance disclosure. This research was conducted with the assistance of AI language models, including Claude (Anthropic), Codex, and other models, which served as coding assistants, documentation aids, experimental design advisors, and analytical tools throughout the project. All scientific decisions, experimental designs, and interpretations were made by the human author. The AI tools were used as assistants in multiple capacities: code implementation, data analysis, literature search, and manuscript preparation.

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A Complete Descriptor Catalog

table 13 lists all 13 descriptor×mechanism combinations explored during the screening phase. Below we provide full mathematical formulations for descriptors not covered in the main text.

Table 13: Complete descriptor catalog. Domain: A = audio, M = MIDI, D = dual. Mechanism: C = concatenation, X = standard cross-attention, R = reverse cross-attention, CM = cross-modal injection.

Arm	Descriptor	Domain	Dims	Mechanism	S_{5ep} (%)	Description
D0	—	—	—	—	60.2	VICReg baseline (no descriptors)
D4	D4	MIDI	4	C	63.6	Local pitch intervals
A4	A4	Audio	8	C	63.6	Octave-band energy dynamics
A7	A7	Audio	12	C	58.8	Rational attractor (just intonation)
A8	A8	Audio	12	C	57.4	Onset-weighted chroma
A9	A9	Audio	12	C	58.8	IDF-weighted rational attractor
D4x	D4	MIDI	4	X	60.0	D4 via standard cross-attention
A4x	A4	Audio	8	X	62.6	A4 via standard cross-attention
A7x	A7	Audio	12	X	62.2	A7 via standard cross-attention
D4r	D4	MIDI	4	R	64.2	D4 via reverse cross-attention
A4r	A4	Audio	8	R	68.6	A4 via reverse cross-attention
d4a4	D4+A4	Dual	4+8	C+C	69.8	D4 concat (MIDI) + A4 concat (audio)
d4a4cm	D4+A4	Dual	4+8	CM	52.4	Cross-modal: A4→MIDI, D4→audio

A.1 A7: Rational Attractor Descriptor

The A7 descriptor projects spectral peak ratios onto 12 just-intonation attractors, capturing the harmonic structure of the signal in a musically meaningful basis.

Just-intonation attractors. We define 12 target ratios corresponding to the just-intonation intervals within one octave (table 14).

Table 14: A7 just-intonation attractors. Log_2 values are octave-folded to $[0, 1)$.

Index	Ratio	$\log_2$	Cents	Musical interval
0	1/1	0.000	0	Unison / Octave
1	16/15	0.093	112	Minor second
2	9/8	0.170	204	Major second
3	6/5	0.263	316	Minor third
4	5/4	0.322	386	Major third
5	4/3	0.415	498	Perfect fourth
6	7/5	0.485	583	Tritone (septimal)
7	3/2	0.585	702	Perfect fifth
8	8/5	0.678	814	Minor sixth
9	5/3	0.737	884	Major sixth
10	7/4	0.807	969	Harmonic seventh
11	15/8	0.907	1088	Major seventh

Algorithm.

1. Compute STFT magnitude $|X[k, t]|$ with $n_{\text{FFT}} = 2048$, hop $= 512$.
2. Extract top-$K = 8$ spectral peaks per frame by magnitude, masking peaks below 50 Hz.
3. Compute all $\binom{8}{2} = 28$ pairwise log-frequency ratios:

$$r_{ij}[t] = \log_2\left(\frac{f_j[t]}{f_i[t]}\right) \bmod 1.0, \quad j > i, \tag{12}$$

   folding into the octave range $[0, 1)$.
4. Soft assignment to attractors via Gaussian kernel ($\sigma = 0.02$):

$$w_{ij,c}[t] = \sqrt{|X_{f_i}[t]| \cdot |X_{f_j}[t]|} \cdot \exp\left(-\frac{(r_{ij}[t] - \mu_c)^2}{2\sigma^2}\right), \tag{13}$$

   where μ_c is the $\log_2$ value of attractor c.

5. Accumulate over all pairs and normalize:

$$A7[t,c] = \frac{\sum_{i<j} w_{ij,c}[t]}{\sum_{c'} \sum_{i<j} w_{ij,c'}[t] + \epsilon}. \tag{14}$$

Output: $A7 \in \mathbb{R}^{B \times T_{\text{stft}} \times 12}$, interpolated to $T = 2400$ for concatenation injection.

A.2 A8: Onset-Weighted Chroma

The A8 descriptor computes a chroma representation weighted by onset strength, emphasizing transient events where note onsets provide the strongest pitch information.

1. Compute STFT magnitude $|X[k,t]|$.
2. Map STFT bins to 12 pitch classes via log-frequency:

$$\text{pc}(k) = \left\lfloor 12 \cdot \log_2\left(\frac{k \cdot f_s / n_{\text{FFT}}}{f_{\text{ref}}}\right)\right\rfloor \bmod 12, \tag{15}$$

   where $f_{\text{ref}} = 32.7\,\text{Hz}$ (C1).
3. Accumulate energy per pitch class: $C[t,c] = \sum_{k:\,\text{pc}(k)=c} |X[k,t]|^2$.
4. Compute spectral flux (onset strength):

$$F[t] = \sum_k \max\bigl(0,\ |X[k,t]| - |X[k,t{-}1]|\bigr). \tag{16}$$

5. Normalize flux to $[0,1]$ and gate chroma: $A8[t,c] = C[t,c] \cdot F[t] / \max(F)$.
6. Normalize per frame: $A8[t,:] \leftarrow A8[t,:]/(\sum_c A8[t,c] + \epsilon)$.

Output: $A8 \in \mathbb{R}^{B \times T_{\text{stft}} \times 12}$.

A.3 A9: IDF-Weighted Rational Attractor

A9 extends A7 with inverse document frequency (IDF) weighting to suppress ubiquitous ratio activations:

1. Compute raw A7 activations $A7_{\text{raw}}[t,c]$ (before normalization).
2. Compute document frequency: $\text{df}_c = |\{t : A7_{\text{raw}}[t,c] > \tau\}|/T$, with $\tau = 0.05$.
3. Compute IDF: $\text{idf}_c = \text{clamp}\bigl(\log(1/(\text{df}_c + 10^{-3})),\ 0,\ 5\bigr)$.
4. Weight and normalize: $A9[t,c] = A7_{\text{raw}}[t,c] \cdot \text{idf}_c / \sum_{c'} A7_{\text{raw}}[t,c'] \cdot \text{idf}_{c'}$.

The IDF weighting suppresses ratio activations that appear in $> 50\%$ of frames (e.g., octave/unison), emphasizing rarer harmonic relationships.

B Architecture Details

B.1 MERTEncoderLite — Full Specification

Table 15: MERTEncoderLite CNN feature extractor.

Layer	In ch.	Out ch.	Kernel	Stride	Padding	Norm	Activation
1	1	512	10	5	0	GN(32)	GELU
2	512	512	3	2	1	GN(32)	GELU
3	512	512	3	2	1	GN(32)	GELU
4	512	1024	3	2	1	GN(32)	GELU

Total downsampling: $5 \times 2 \times 2 \times 2 = 40\times$. Input $96{,}000 \rightarrow$ output $2{,}400$ frames.

Table 16: Audio Transformer encoder specification.

Hyperparameter	Value
Number of layers	4
Model dimension (d_{model})	1024
Number of heads	8
Head dimension (d_k)	128
Feed-forward dimension (d_{ff})	4096
Activation	GELU
Dropout	0.1
Positional embeddings	Learnable, max 6000
Pooling	Mean

Table 17: MIDI event embedding dimensions.

Feature	Vocabulary	Embedding dim	Fraction of d_{model}
Pitch	128	256	$d/2$
Velocity	128	128	$d/4$
Duration	32	128	$d/4$
Concatenated		512	d
+ Linear proj. + LayerNorm		512	

B.2 MIDI Encoder — Full Specification

B.3 Parameter Count Breakdown

C Gate 4.4: Architecture Family Results

C.1 FiLM Conditioning

Feature-wise Linear Modulation [Perez et al., 2018] modulates Transformer layer activations:

$$F'_\ell = (1 + \gamma_\ell) \odot F_\ell + \beta_\ell, \tag{17}$$

where $\gamma_\ell, \beta_\ell \in \mathbb{R}^{d_{\text{model}}}$ are generated from the descriptor via a 2-layer MLP:

$$[\gamma_\ell, \beta_\ell] = W_2 \; \mathrm{ReLU}(W_1 \, \bar{d} + b_1) + b_2, \tag{18}$$

with $\bar{d} = \mathrm{MeanPool}(d)$ being the temporally averaged descriptor.

Three FiLM variants were tested: `film-a4` (A4 only), `film-d4` (D4 only), `film-dual` (both). None improved over D0 (best: 59.4%, D0: 60.2%).

C.2 Mixture-of-Experts

We implemented a sparsely-gated MoE layer [Shazeer et al., 2017] replacing the feed-forward sub-layer in the final Transformer block:

$$\mathrm{MoE}(x) = \sum_{i=1}^{E} g_i(x) \cdot \mathrm{FFN}_i(x), \quad g(x) = \mathrm{TopK}\big(\,\mathrm{softmax}(W_g \, x),\; k\big), \tag{19}$$

with $E = 4$ or 8 experts and $k = 2$. A load-balance loss $\mathcal{L}_{\text{bal}}$ and entropy regularizer $\mathcal{L}_{\text{ent}}$ encourage uniform expert utilization.

Five MoE variants were tested (`moe-a4`, `moe-dual`, `moe-a4-v2` through `v4`) with varying numbers of experts, top-k values, and gating architectures. Best: 60.2% (moe-a4-v2), matching D0 exactly.

Table 18: MIDI Transformer encoder specification.

Hyperparameter	Value
Number of layers	4
Model dimension (d_{model})	512
Number of heads	8
Head dimension (d_k)	64
Feed-forward dimension (d_{ff})	2048
Activation	GELU
Dropout	0.1
Norm order	Pre-norm
Positional encoding	Sinusoidal, max 2048
Pooling	Masked mean
Output norm	LayerNorm(512)

Table 19: Parameter count breakdown for each Gate 5B model.

Component	D0	d4a4	a4r	d4-a4r
Audio CNN	3.2M	3.2M	3.2M	3.2M
Audio Transformer	54.5M	54.5M	54.5M	54.5M
Audio pos. embeddings	6.1M	6.1M	6.1M	6.1M
MIDI embeddings	1.2M	1.2M	1.2M	1.2M
MIDI Transformer	10.5M	10.5M	10.5M	10.5M
Audio projection	0.8M	0.8M	0.8M	0.8M
MIDI projection	0.4M	0.4M	0.4M	0.4M
A4 injection	—	0.01M	—	—
A4 reverse cross-att	—	—	4.2M	4.2M
D4 injection	—	0.01M	—	0.01M
A4 concat proj.	—	0.01M	—	—
Total	**74.2M**	**75.5M**	**78.6M**	**78.9M**

C.3 Third Tower

The Third Tower architecture introduces a separate descriptor encoder whose representations are aligned with both audio and MIDI via auxiliary VICReg losses:

$$\mathcal{L}_{\text{t3}} = \mathcal{L}_{\text{VICReg}}(z_a, z_m) + \alpha\, \mathcal{L}_{\text{VICReg}}(z_a, z_d) + \beta\, \mathcal{L}_{\text{VICReg}}(z_m, z_d), \tag{20}$$

where z_d is the descriptor tower output.

Three variants were tested:

- `t3-wt` (weighted loss, $\alpha = \beta = 0.5$): $S = 67.6\%$ ($+7.4$ pp). Best of the family.
- `t3-tri` (triplet-style): $S = 65.0\%$ ($+4.8$ pp).
- `t3-anc` (anchor, descriptor as anchor for both): $S = 42.2\%$ (-18.0 pp). Catastrophic.

The t3-wt variant was trained to 50 epochs, achieving $S = 81.2\%$—competitive with direct injection methods.

D Training Hyperparameters

No data augmentation. We deliberately omit data augmentation (pitch shift, time stretch, noise injection) to isolate the effect of descriptor injection from augmentation-based regularization. This conservative choice means our results represent a lower bound on achievable performance.

Scheduler variants explored.

- **Cosine**: Standard cosine annealing [Loshchilov and Hutter, 2017] from base LR to 0 over T_{ref} epochs.

Table 20: Complete training hyperparameters for Gate 5B canonical models.

Parameter	D0	d4a4	a4r	d4-a4r
Optimizer	AdamW (β_1=0.9, β_2=0.999)			
Weight decay	0.01			
Batch size	16			
Total epochs	60	60	30	30
Best epoch	50	50	29	30
Warmup steps	500			
Schedule	Cosine-tail	Cosine-60	Cosine-30	Cosine-30
LR floor	0.10	—	—	—
LR tail end	0.02	—	—	—
Cosine ref epochs	30	60	30	30
VICReg λ_{inv}	10			
VICReg λ_{var}	10			
VICReg λ_{cov}	1			
Segment length	4 s (96,000 samples at 24 kHz)			
Segment hop	1 s			
Max MIDI notes	2048			
Duration buckets	32 (log-spaced, 0.05–4 s)			
DataLoader workers	8			
Pin memory	True			
Prefetch factor	2			
Data augmentation	None (deliberate)			

- **Cosine-tail**: Cosine annealing with a floor (10% base LR) and linear tail ($\rightarrow 2\%$) for epochs beyond T_{ref}.
- **Trapezoidal**: Linear warmup $\rightarrow$ constant hold $\rightarrow$ linear decay.

Evaluation configuration (canonical, pinned).

- Pool size: 256 random items
- Number of queries: 500
- Hard negatives: 64 (same piece, different time)
- Semi-hard negatives: 32 (same composer, different piece)
- Random seed: 42
- Embedding dimension: 256

E Additional Visualizations

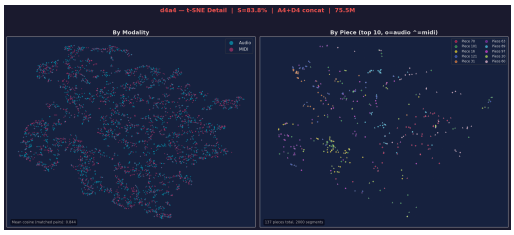
E.1 t-SNE and UMAP Embedding Spaces

figs. 9 and 10 show t-SNE and UMAP projections of 2000 audio (blue) and MIDI (red) embeddings for each model. Matched pairs are connected by light gray lines.

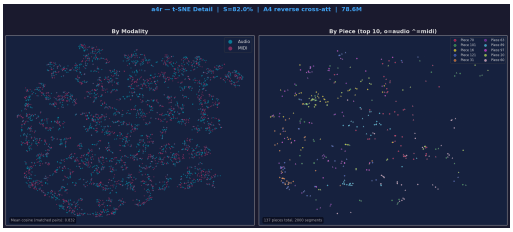
E.2 Alignment Cosine Distributions

The distribution of cosine similarities between matched audio–MIDI embedding pairs provides a complementary view to the retrieval metrics. Higher mean cosine with lower variance indicates more consistent alignment.

(a) D0 — t-SNE



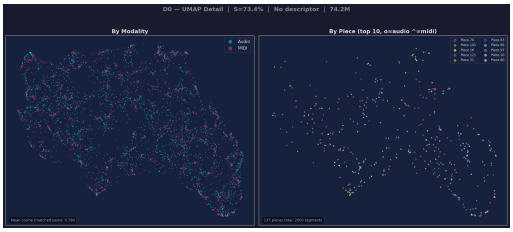
(b) d4a4 — t-SNE



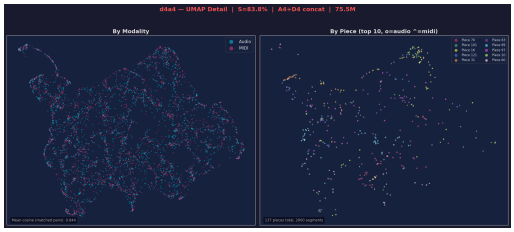
(c) a4r — t-SNE

(d) d4-a4r — t-SNE

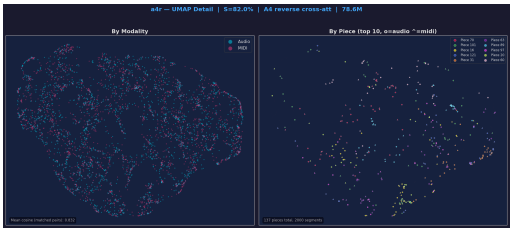
Figure 9: t-SNE embeddings for the 4 canonical models from the seed-42 checkpoint. Blue: audio; red: MIDI. Lines connect matched pairs. Figures are rendered from Test 10 exported PNGs.



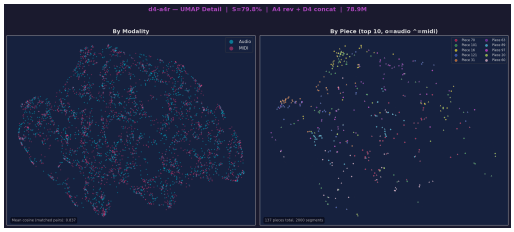
(a) D0 — UMAP



(b) d4a4 — UMAP



(c) a4r — UMAP



(d) d4-a4r — UMAP

Figure 10: UMAP embeddings for the 4 canonical models from the seed-42 checkpoint (same split and sample count as t-SNE).

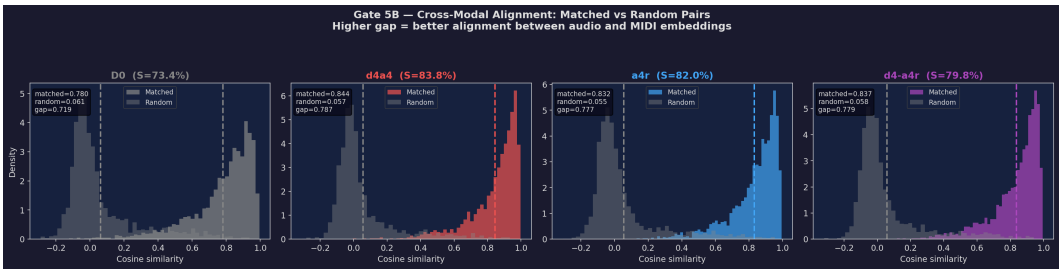


Figure 11: Distribution of cosine similarity for matched audio–MIDI pairs across canonical models from the seed-42 checkpoint (Test 10).